

YAWEN GUO

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EDUCATION

University of Washington (UW)

M.S. Information Management

09/2023 - 08/2024

B.S. Informatics (Software Engineering)

09/2020 - 06/2023

SKILLS

Programming Languages: TypeScript, JavaScript, Python, Java, R, SQL

Technologies: React, Next.js, HTML5, CSS3, Tailwind, Bootstrap, Sass, Chart.js, D3.js, Node.js, REST APIs, CLI Tools, Firebase, MongoDB, Postgres, Git, Docker, Vercel, Microsoft Azure, AWS, WebSockets, Jest, Agile Dev

PROFESSIONAL EXPERIENCE

Software Engineer, Columbia Crossings LLC

02/2025 - Present

- Design and maintain a suite of production web applications using React, TypeScript, and Node.js to support end-to-end marina operations, including customer reservation intake for boat moorage and storage, document upload and verification, and internal staff tooling for reservation processing.
- Lead the transition from multiple single-tenant sites to a unified multi-tenant architecture using Payload CMS, defining data models, access control, and deployment strategy to support multiple marinas and business units within a single platform.
- Integrate real-time services with Firebase and WebSockets to synchronize reservation status and user activity across client-facing and internal systems.
- Develop and maintain REST APIs with schema validation, error handling, and logging to ensure reliable data flow between front-end applications, the CMS, and third-party services.
- Create internal developer tooling and CLI scripts to automate environment setup, local emulation, and deployment workflows, improving developer productivity and reducing operational friction.
- Profile and optimize high-traffic pages and file-upload pipelines using Lighthouse and runtime diagnostics, reducing latency and improving system responsiveness.
- Document system architecture, multi-tenant data isolation strategy, and operational runbooks to support onboarding, scalability, and long-term maintainability.

Website Developer (Internship), UW University Marketing and Communications

02/2024 - 08/2024

- Rebuilt an internal KPI dashboard by replacing static visualization assets with interactive, data-driven charts using JavaScript, React, and Chart.js, enabling dynamic filtering, drill-down, and real-time insight exploration.
- Maintained and extended a custom WordPress theme using PHP, JavaScript, and CSS, reviewing legacy code, fixing bugs, and building new UI features to support university marketing and content management needs.

Software Engineer (Internship), AIMHealth Innovation Lab

09/2023 - 12/2024

- Built a healthcare web application using Next.js and Tailwind CSS that transforms free-text symptom input into structured, readable clinical insights for non-technical users.
- Designed and implemented the Express + TypeScript API backing the UI, defining REST endpoints for session management and chat workflows that integrate the OpenAI API for AI-generated responses.
- Integrated Firebase Admin authentication, verifying ID tokens at login and enforcing session-cookie-based authorization across protected routes.
- Implemented request validation, error handling, and middleware-based guards to ensure reliable and predictable API behavior under real user input.
- Containerized the backend service with Docker, compiling TypeScript during image build and running the API via Docker Compose for consistent local and server environments.
- Partnered with clinicians and UX designers to run usability testing, iterating on interaction flows, messaging, and output structure to improve clarity and trust in clinical contexts.

PROJECTS

Front-end Software Engineer, Nutritional Food Guidance for Food Banks

03/2023 - 06/2023

- Built a client-side search and filtering system in React and Bootstrap to index and query a large catalog of food items, supporting multi-criteria filtering with responsive updates and low-latency interaction.
- Designed component state and data flow to handle dynamic queries, pagination, and real-time UI updates without full page reloads, improving performance and user experience on low-bandwidth devices.
- Refactored UI components for modularity and reuse, and added unit tests with Jest to validate filtering logic, state transitions, and edge cases, reducing regressions during iteration.

Full-stack Software Engineer, Travel Plan Customizing Tool

03/2022 - 06/2022

- Developed a small web application integrating third-party APIs to generate personalized travel plans, focusing on frontend data handling and user-driven workflows.